SEQUENCES LADDER 2 OF 9

# Second differences and quadratic rules

A constant second difference $D$ signals a quadratic rule $u_n = \frac{D}{2}n^2 + bn + c$.

## 1 / THE RULE AND THE PICTURE

### If the second differences all equal $D$ then $a = D/2$

| | | | | | |
|---|---|---|---|---|---|
| $u_n$ | 2 | 6 | 12 | 20 | 30 |
| $\Delta$ | 4 | 6 | 8 | 10 | |
| $\Delta^2$ | 2 | 2 | 2 | | |

The first differences $4, 6, 8, 10$ are not constant, but their own differences, the second differences, are all 2, and

$$\Delta^2 u_n = D \implies u_n = \tfrac{D}{2}\,n^2 + bn + c$$

Here $D = 2$, so the rule begins with $1 \cdot n^2$, and the remaining part is found by subtracting $n^2$ from every term.

## 2 / THE RECIPE

### Subtract the square part and read off the straight line left over

For $2, 6, 12, 20, 30$ the second difference is 2, so $a = 1$, and subtracting $n^2 = 1, 4, 9, 16, 25$ leaves $1, 2, 3, 4, 5$, which is $n$, so $u_n = n^2 + n$.
For $5, 12, 23, 38, 57$ the first differences are $7, 11, 15, 19$ and the second differences are $4$, so $a = 2$, and subtracting $2n^2 = 2, 8, 18, 32, 50$ leaves $3, 4, 5, 6, 7 = n + 2$, so $u_n = 2n^2 + n + 2$.

## 3 / WHY $D = 2a$

### Differences of a quadratic are a straight line

| | formula | meaning |
|---|---|---|
| $u_n = an^2 + bn + c$ | $an^2 + bn + c$ | |
| first difference | $a(2n + 1) + b = 2an + (a + b)$ | grows by $2a$ each step |
| second difference | $2a$ | constant, so $D = 2a$ |

The step from $an^2$ to $a(n + 1)^2$ is $a(2n + 1)$, which itself grows by $2a$ each step, so the second difference is $2a$ whatever $b$ and $c$ are.

## 4 / FAMOUS QUADRATIC SEQUENCES

### Triangular, square and pentagonal numbers

The triangular numbers $1, 3, 6, 10, 15$ have $D = 1$, so $a = \frac{1}{2}$, and subtracting $\frac{1}{2}n^2$ leaves $\frac{1}{2}n$, giving $\frac{n(n+1)}{2}$.
The squares $1, 4, 9, 16, 25$ have $D = 2$ and rule $n^2$. The pentagonal numbers $1, 5, 12, 22, 35$ have first differences $4, 7, 10, 13$ and $D = 3$, so $a = \frac{3}{2}$ and subtracting $\frac{3}{2}n^2$ leaves $-\frac{1}{2}n$, giving $\frac{3n^2 - n}{2}$.

## 5 / WHAT THE SECOND DIFFERENCE MEASURES

### How fast the gradient is changing

The first differences are the gradients between neighbouring points, and the second difference says how much the gradient grows each step, which for a quadratic is the same everywhere, $2a$.
This is the discrete form of the curvature of a parabola, and in calculus it becomes the second derivative.

## 6 / TRY THESE

### Use the recipe

**(a)** For $4, 9, 16, 25, 36$ the second difference is 2, so $n^2 + 2n + 1 = (n + 1)^2$. **(b)** For $0, 3, 8, 15, 24$ it is 2, and subtracting $n^2$ leaves $-1, -1, -1$, so $n^2 - 1$. **(c)**
For $3, 5, 9, 15, 23$ the differences $2, 4, 6, 8$ give $D = 2$, and subtracting $n^2$ leaves $2, 1, 0, -1, -2 = 3 - n$, so $n^2 - n + 3$.

**ASIDE / TWO LEVELS DEEP**

A quadratic needs three terms to fix its three constants and a fourth to confirm that the second differences really are constant.

**LINKS TO OTHER SHEETS**

The previous sheet gives first differences; the next sheet goes to third differences and cubics.

**Convention.** $\Delta^2 u_n = u_{n+2} - 2u_{n+1} + u_n$ is the second difference; $D$ is its constant value; $a = D/2$ is the coefficient of $n^2$.